End-to-End Azure Data Engineering and Analytics Pipeline Using ADLS, Synapse, and Power BI

1. Project Overview

This project demonstrates the implementation of a cloud-based end-to-end data engineering and analytics solution using Microsoft Azure services. The solution ingests raw booking and property data into Azure Data Lake Storage (ADLS), processes and transforms the data using Apache Spark in Azure Synapse Analytics, loads analytics-ready data into a Synapse Dedicated SQL Pool, and visualizes insights using Power BI dashboards.

The project simulates a real-world analytics platform for customer booking and property analysis.

2. Architecture Overview

Architecture Flow

Raw Data Files
↓
Azure Data Lake Storage Gen2 (ADLS)
↓
Azure Synapse Analytics – Spark Notebook
↓
Delta Lake Tables
↓
Data Cleansing and Transformation
↓
Synapse Dedicated SQL Pool
↓
Power BI Dashboard and Reporting

3. Technologies Used

ADLS Gen 2, Azure Synapse Analytics, Apache Spark (PySpark), Delta Lake, Synapse Dedicated SQL pool,JDBC Connector, Power BI

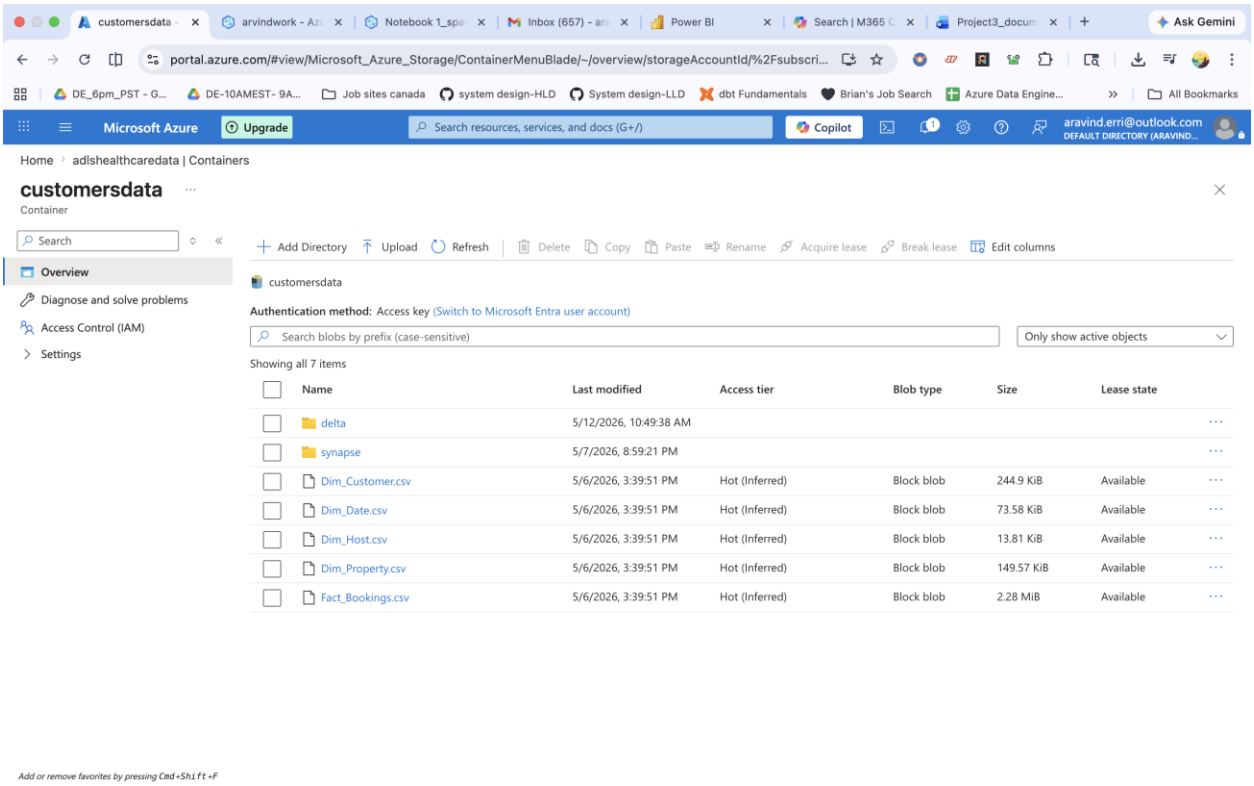
4. Data Ingestion into ADLS

Objective

Load raw customer booking and property datasets into Azure Data Lake Storage Gen2.

Steps Performed

1. Created an Azure Data Lake Storage Gen2 account
2. Created raw container
3. Uploaded source datasets into ADLS
4. Connected Synapse workspace to ADLS
5. The source data was fact and dimension tables, from Kaggle



5. Data Processing Using Synapse Spark

Objective

Transform raw datasets into analytics-ready datasets using PySpark.

Activities Performed

- Loaded source files from ADLS
- Cleaned and standardized columns
- Performed joins and transformations
- Created analytics DataFrames:
    - df_customer_analytics
    - df_property_analytics
- Stored transformed data in Delta format

6. Delta Table Creation

Objective

Store curated analytics data in Delta Lake format for reliability and optimized querying.

Delta Tables Created

Customer_analytics

Property_analytics

Benefits of Delta Lake

- ACID transactions
- Schema enforcement
- Improved query performance
- Time travel support

7. Synapse Dedicated SQL Pool Integration

Objective

Load analytics-ready data from Spark DataFrames into Synapse Dedicated SQL Pool for enterprise reporting and BI consumption.

8. JDBC Connection Configuration

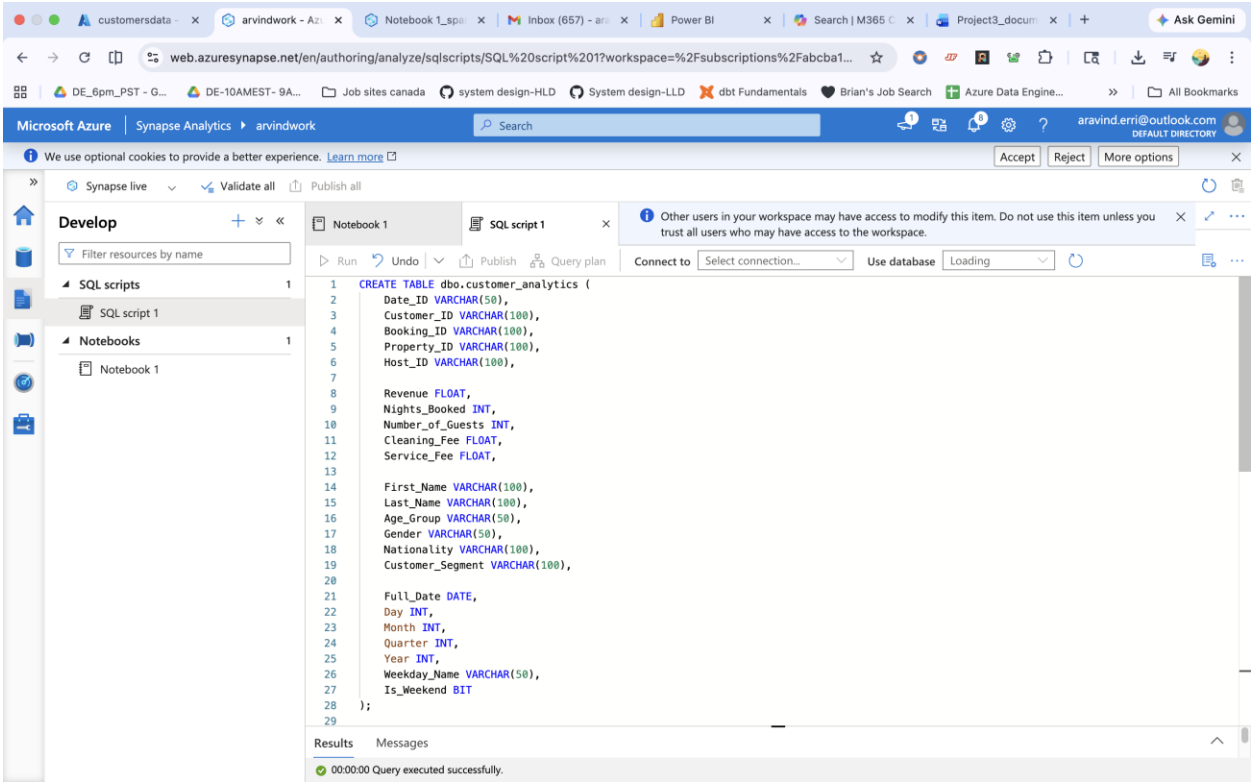
```
jdbc_url = "jdbc:sqlserver://arvindwork.sql.azuresynapse.net:1433;database=dedicatedsqlpool"


connection_properties = {
    "user": "sqladminuser",
    "password": "<password>",
    "driver": "com.microsoft.sqlserver.jdbc.SQLServerDriver"
}
```

9. SQL Table Creation in Dedicated SQL Pool

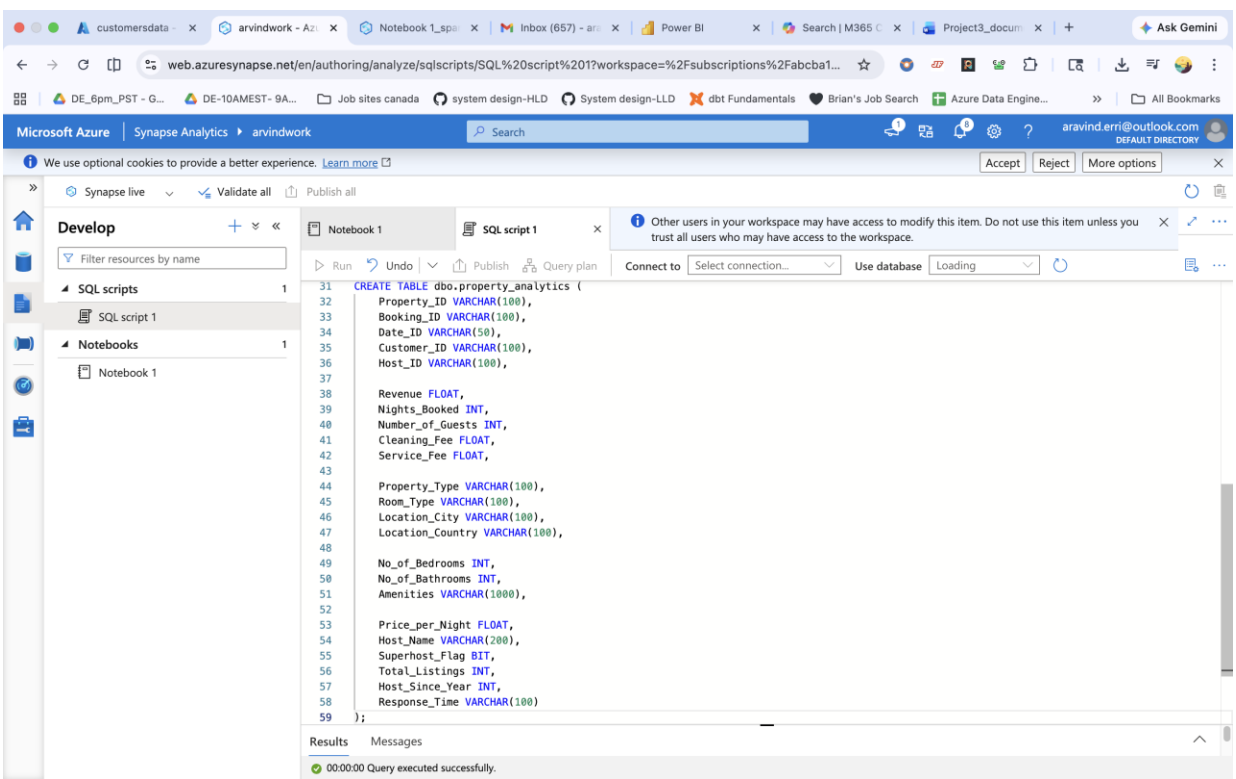
Objective

Pre-create SQL tables before loading data to avoid schema mismatch and columnstore index issues.

1.customer_analytics table



Property_analytics table



10. Loading Data into Dedicated SQL Pool

Spark Append Write

Customer_analytics

```
df_customer_analytics.write \
    .mode("append") \
    .jdbc(url=jdbc_url,
          table="dbo.customer_analytics",
          properties=connection_properties)
```

property_analytics

```
df_property_analytics.write \
    .mode("append") \
    .jdbc(url=jdbc_url,
          table="dbo.property_analytics",
          properties=connection_properties)
```

12. Data Validation in SQL Pool

Validate Row Count

```
SELECT COUNT(*)
FROM dbo.customer_analytics;
```

Validate Sample Records

```
SELECT TOP 10 *
FROM dbo.customer_analytics;
```

13. Power BI Integration

Objective

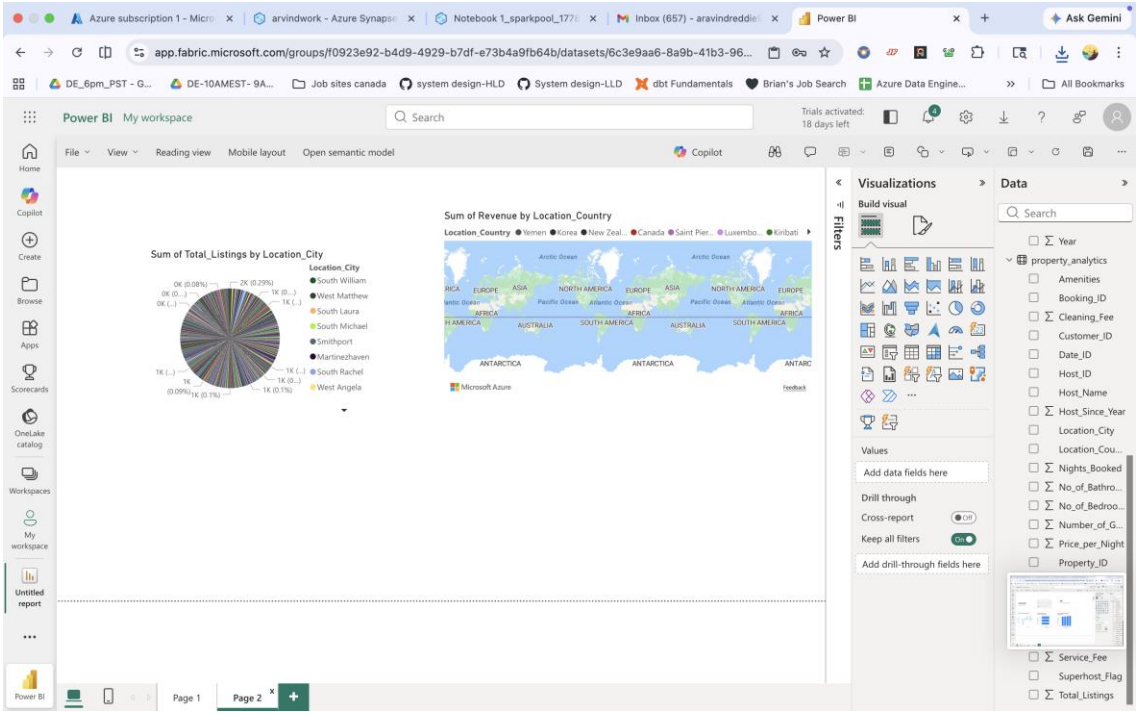
Build interactive dashboards using Synapse Dedicated SQL Pool as the data source.

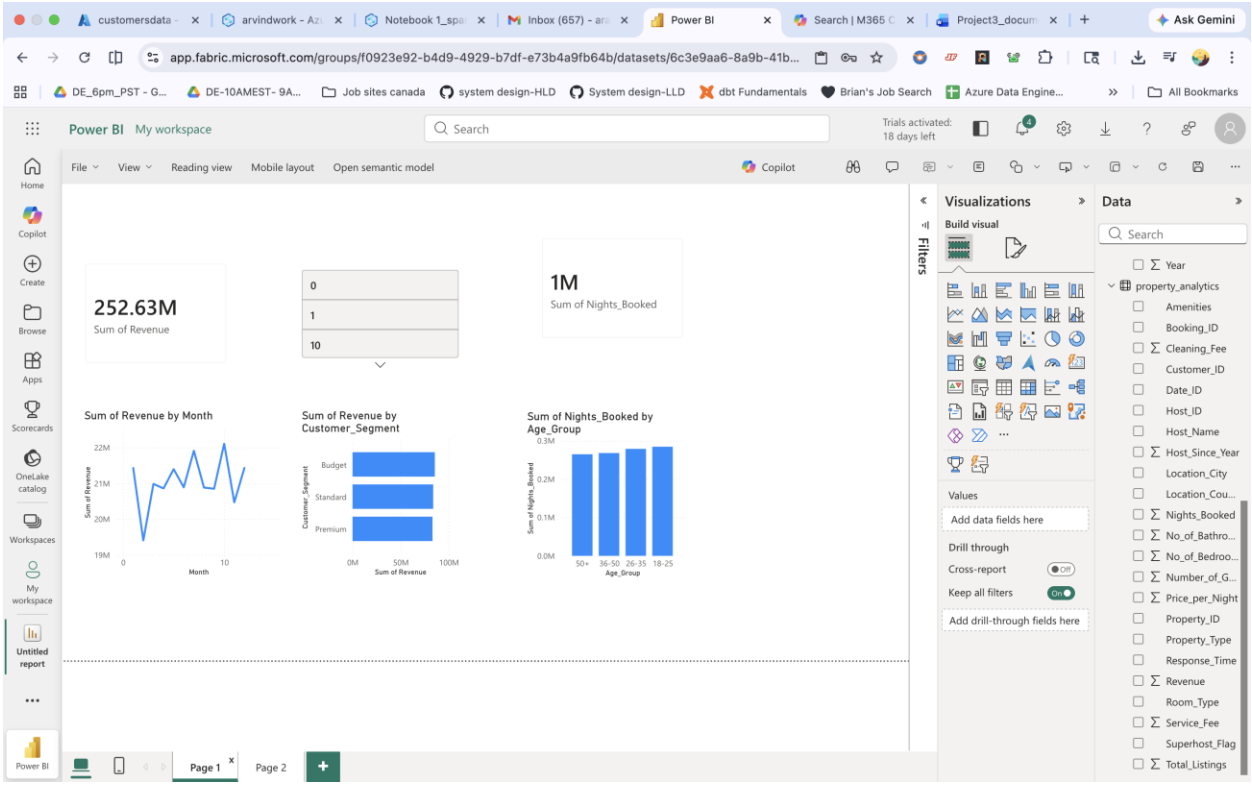
Steps Performed

1. Connected Power BI to Synapse Dedicated SQL Pool
2. Imported analytics tables
3. Created measures and relationships
4. Designed visual dashboards

Dashboard KPIs

- Total Revenue
- Total Bookings
- Revenue by Country
- Revenue by Property Type
- Customer Segmentation
- Weekend vs Weekday Analysis
- Host Performance Metrics





Transformation Overview:

First, based on the 1 fact and 4-dimension tables,i did few transformations like join ,groupby and created 2 data frames ,and write into 2 delta tables and performed filter condition on two deltatables and upsert that data to deltatables and merged the data to delta tables

df1---> deltatable 1

df2--->deltatable 2

df1_filter 1----upsert---------->deltatable 1

df2_filter2-------upsert--------->deltatable 2

df1_filter 1----merge---------->deltatable 1

df2_filter2-------merge--------->deltatable 2

17. Conclusion

This project successfully implemented an end-to-end Azure analytics pipeline using Azure Data Lake Storage, Azure Synapse Analytics, Delta Lake, Synapse Dedicated SQL Pool, and Power BI.

The project demonstrated modern cloud-based data engineering practices for ingesting, transforming, warehousing, and visualizing analytics data at scale. It also provided practical exposure to real-world challenges such as JDBC connectivity issues, schema mismatches, duplicate columns, and performance bottlenecks, along with their corresponding resolutions.

The final architecture supports scalable enterprise analytics and business intelligence reporting using Microsoft Azure services.